



Air Conditioning and
Refrigeration Center

Water-Cooled Chiller for High Efficiency Edge Compute Data Center Facility

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Course Project: IRCC 2026 — Project #3

150 kW water-cooled propane chiller · Champaign, IL

Cycle design, component selection & annual performance

Compute Demand Is Climbing — and Cooling Has to Keep Up

- **Demand is growing fast** — AI, cloud and edge compute keep adding server halls, and every new rack is heat
- **Every IT watt needs a cooling path out** — heat rejection is mission-critical, not overhead
- Cooling is the **largest non-IT energy consumer** — its efficiency decides the facility's **PUE**
- IT runs **24/7/365** — the cooling plant can never stop
- **Efficient cooling frees grid power for compute** — the point of this design



Image: gridx.com

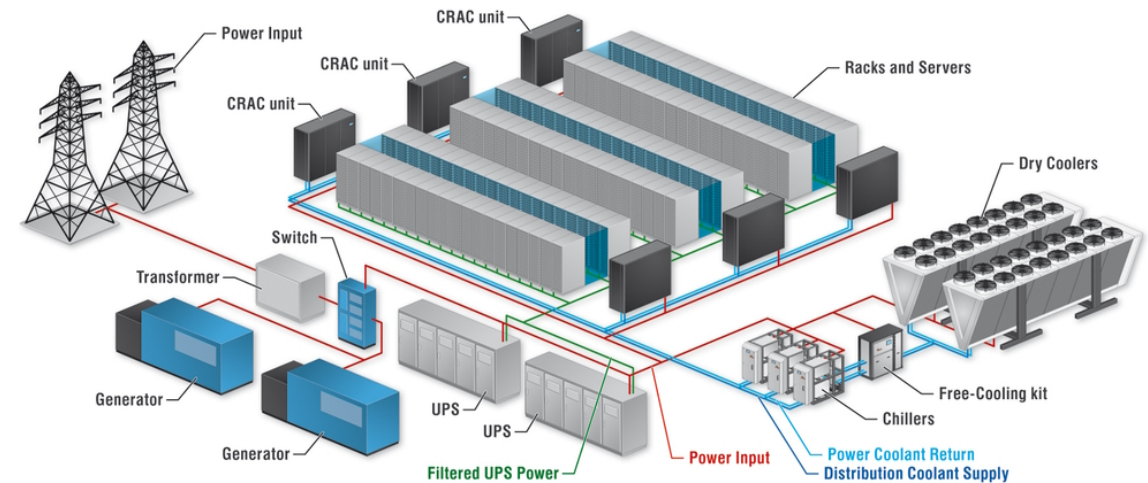
The Design Task



Deliverables & Hard Targets

- Full cycle design with working-fluid comparison
- Selection & sizing of every major component
- Seasonal performance — **COP, IPLV, PUE**
- BOM · p-h chart · P&ID · safety · controls

PARAMETER	REQUIREMENT
Maximum cooling load	150 kW (≈ 42 RT), sensible IT
Minimum load / turndown	30 kW — 5:1
Chilled water supply / return	15 / 21 °C (ASHRAE TC 9.9 W17)
White space	8 × 12 m · 25 racks · 24/7/365
Design ambient	35 °C DB · TMY3 Champaign
Full-load COP	≥ 3.5 at the AHRI point (ASHRAE 90.1)
IPLV.IP	≥ 5.0 (AHRI 550/590-2023)
Economizer	active ≤ 10 °C · full ≤ 4 °C OAT (§6.5.1)
Refrigerant	GWP < 150 · A2L/A3 + charge analysis

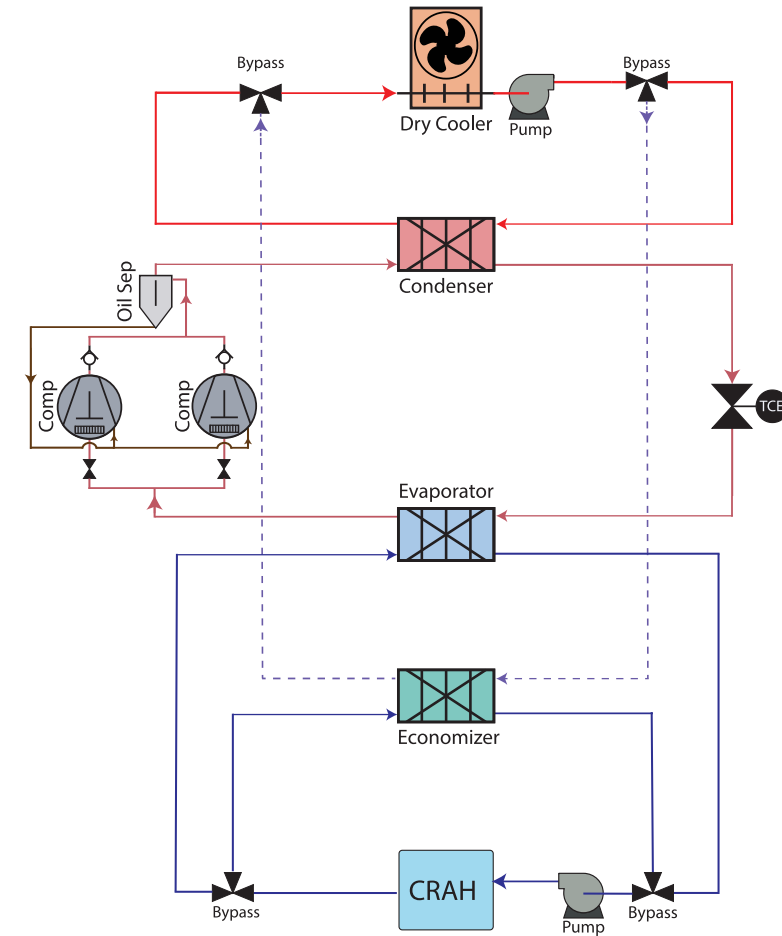


Proposed Design



Three Loops in Series, Economizer in Parallel

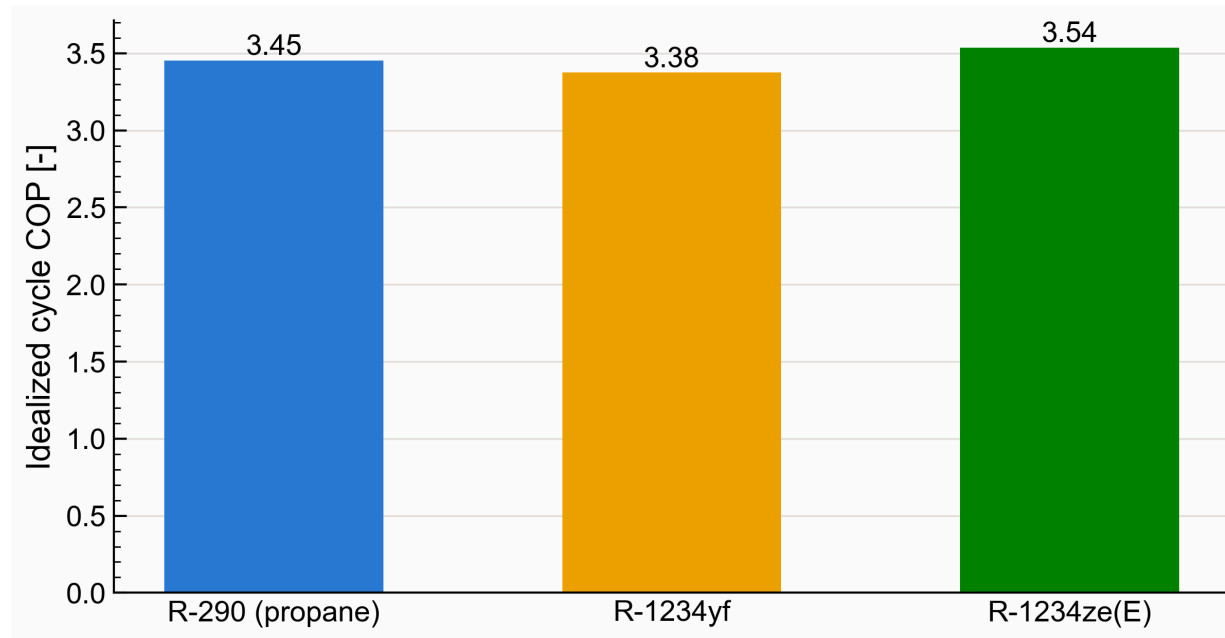
- Three loops in series: **chilled water** → **R-290** → **glycol** → **air**
- The 30% propylene-glycol loop keeps flammable A3 propane **out of the data hall**
- A **waterside economizer** in parallel with the chiller enables compressor-off free cooling
- Two parallel VFD compressors on **one shared refrigerant circuit**



System P&ID — refrigerant, glycol and chilled-water loops

COP of the Screened Low-GWP Candidates

- **R-290 moves ~1.7× more cooling per unit swept volume** (3,270 vs 1,880 kJ/m³) — R-1234ze(E) would need a ~1.7× larger compressor for the same 150 kW.
- The penalty is **size, not efficiency** — R-1234ze(E)'s COP edge is real but tiny (~2%); its low suction pressure just demands a bigger machine.
- Bigger compressor + larger suction piping means **more cost and floor space** — a poor trade in a space-constrained edge hall.



Cycle COP across screened refrigerant candidates

Idealized single-stage cycle · 7 / 52 °C · 8 K superheat · 5 K subcool · $\eta_{isen} = 0.70$

SELECTED WORKING FLUID

R-290

- GWP 3
- Zero ODP
- High volumetric capacity
- Low cost
- Widely available

A future-proof natural refrigerant

Evaporator — Brazed-Plate HX



Evaporator — two-zone model

- **Two zones** — boiling (Han) + superheat (Blasius), split by enthalpy; the superheat tail sets the **suction state**.
- 150 kW across $\Delta T_{lm} = 7.6$ K needs 8.2 m² at **$U = 2,400$ W/m²K** (double-wall derated): **≈84 plates**, ~20% margin.
- $R-290 \cdot N_{cp} = 38 \cdot G_c \approx 25.3$ kg/m²s — inside Han's 13–34 kg/m²s band.

ΔP — REFRIGERANT SIDE (BOILING ZONE)

$$\Delta P_{fr} = f \cdot (L_v N_{cp} / D_h) \cdot G_{Eq}^2 / \rho_f$$
$$f = Ge_3 \cdot Re_{Eq}^{Ge_4} \quad G_{Eq} = G_c [(1 - x) + x(\rho_f / \rho_g)^{1/2}] \quad G_c = \dot{m} / (N_{cp} b L_w)$$

ΔP — WATER SIDE

$$\Delta P = 2f \cdot (L_v / D_h) \cdot G^2 / \rho \quad \text{— Martin (1996), single-phase}$$

$$5.973 \text{ kg/s} \cdot G = 545 \text{ kg/m}^2\text{s} \cdot Re = 2,072 \cdot f = 0.109 \rightarrow \Delta P = 9.75 \text{ kPa}$$

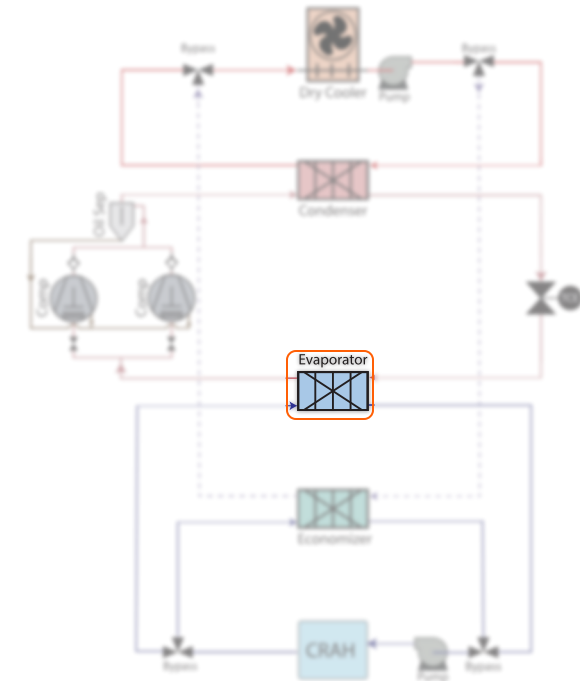
WHERE F COMES FROM — HAN (2003), EVAPORATION SET

$$Ge_3 = 64,710 \cdot (p_{co} / D_h)^{-5.27} \cdot (\pi/2 - \beta)^{-3.03}$$

$$\text{At } \beta = 30^\circ, p_{co} = 5 \text{ mm}, D_h = 4 \text{ mm} \rightarrow Ge_3 \approx 17,361, Ge_4 \approx -1.120$$

D_h 4 mm · β 30° · p_{co} 5 mm · b 2.34 mm (derived) · L_w 200 mm · L_v 0.5 m — from HX geometry

SELECTED MODEL → Kelvion HP DW 500H · double-wall · ≈84 plates



Evaporator — location in the system



Kelvion double-wall brazed-plate HX

Condenser — Brazen-Plate HX



Condenser — three-zone model

- Three zones by **enthalpy, not length** — 14.0 / **81.5** / 4.5%: one LMTD suffices; subcool gives the EEV **5 K liquid** — no receiver.
- Design point: **182 kW** at $t_c = 45\text{ °C}$, glycol 38 → 42 °C; $U = 2,000\text{ W/m}^2\text{K}$ → 12.2 m²: **≈104 plates**.

ΔP — REFRIGERANT SIDE (CONDENSING ZONE)

$\Delta P_{fr} = f \cdot (L_v N / D_h) \cdot G_{Eq}^2 / \rho_f$ — integrated over 200 quality steps, $x = 0 \rightarrow 1$
(np.trapezoid)

Desuperheat + subcool zones (Blasius): $\Delta P = 2f \cdot (G^2 / \rho) \cdot L_{zone} / D_h$

$N = 2$ is Han's rig constant, not $N_{cp} = 47$ (else 23.5× over) · $G_c \approx 25.1$ — inside Han's 13–34 band

ΔP — GLYCOL SIDE

$\Delta P = 2f \cdot (L_v / D_h) \cdot G^2 / \rho$ — Martin (1996)

11.64 kg/s · $Re = 2,703$ (turbulent branch) · $f = 0.107 \rightarrow \Delta P = 35.91\text{ kPa}$

4× the evaporator water loss at equal area — ~2× flow, denser fluid, $\Delta P \propto G^2$

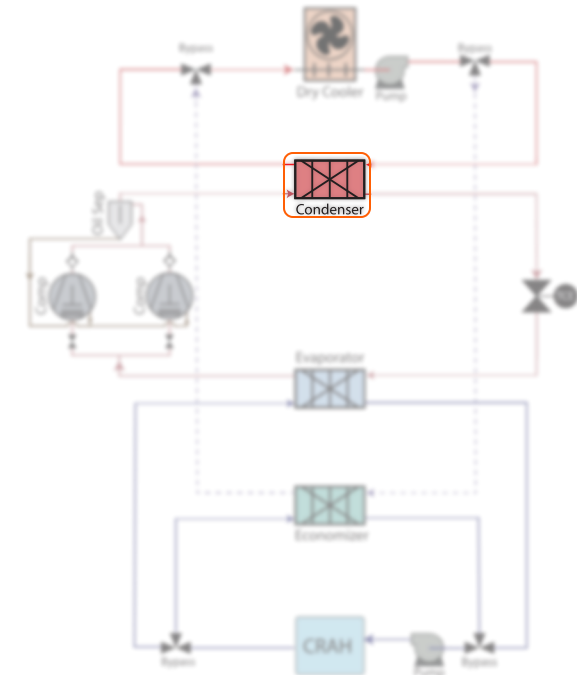
WHERE F COMES FROM — HAN (2003), CONDENSATION SET

$Ge_3 = 3,521.1 \cdot (p_{co} / D_h)^{+4.17} \cdot (\pi/2 - \beta)^{-7.75}$

Same β and p_{co} as the evaporator → $Ge_3 \approx 6,246$, $Ge_4 \approx -0.985$

Sign flips (+4.17 vs -5.27): film growth vs nucleation — $f \approx 2.8\times$ lower

SELECTED MODEL → **Kelvion HP DW 500H · double-wall · ≈104 plates**



Condenser — location in the system



Kelvion double-wall brazed-plate HX

Economizer Selection and Modeling

- **Hardest duty point:** 100% of 150 kW at **4 °C OAT** (ASHRAE 90.1 §6.5.1)
- LMTD at that point → area → **≈49 plates** · single-wall — **no refrigerant contact**
- **Payoff:** 2,542 h/yr of compressor-off free cooling — 29% of the year

ΔP — BOTH SIDES, SAME EQUATION

$$\Delta P = 2f \cdot (L_v / D_h) \cdot G^2 / \rho \quad G = \dot{m} / (N_{cp} b L_w)$$

Water: $Re = 1,617 \cdot f = 0.105 \rightarrow 6.67 \text{ kPa}$ | Glycol: $Re = 456 \cdot f = 0.142 \rightarrow 10.43 \text{ kPa}$

Both on the laminar branch — the condenser's glycol at $Re = 2,703$ is turbulent

WHERE F COMES FROM — MARTIN (1996)

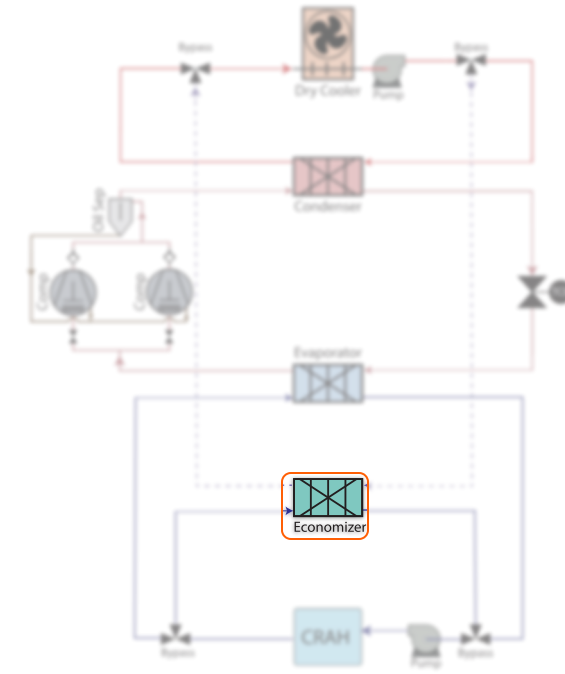
$$\frac{1}{\sqrt{f}} = \frac{\cos \varphi}{\sqrt{0.045 \tan \varphi + 0.09 \sin \varphi + f_0 / \cos \varphi}} + \frac{1 - \cos \varphi}{\sqrt{3.8 f_1}}$$

$$Re < 2000: \quad f_0 = 16 / Re, \quad f_1 = 149 / Re + 0.9625$$

φ is the chevron angle, used directly as $\beta = 30^\circ$ — no 90° – β conversion

One shared `_martin_dp()` serves every single-phase stream in the plant

SELECTED MODEL → **Kelvion GB series · single-wall · ≈49 plates**



Economizer — location in the system



Kelvion plate heat exchanger, glycol → chilled water

Compressor Selection and Modeling

- Two parallel 75 kW units on **one shared circuit** — redundancy + staging
- Modeled by the manufacturer's **AHRI-540 polynomial map** — reproduces Bitzer's rating (74.9 kW, 22.74 kW, COP 3.55) exactly
- **VFDs** trim full-load speed to $\approx 75\%$ for exactly 150 kW, and carry the 5:1 turndown without hot-gas bypass
- **Modeling:** three coefficient sets — capacity, power, mass flow — each evaluated at (t_o, t_c) ; **solve_with_vfd()** finds the speed that delivers the load exactly

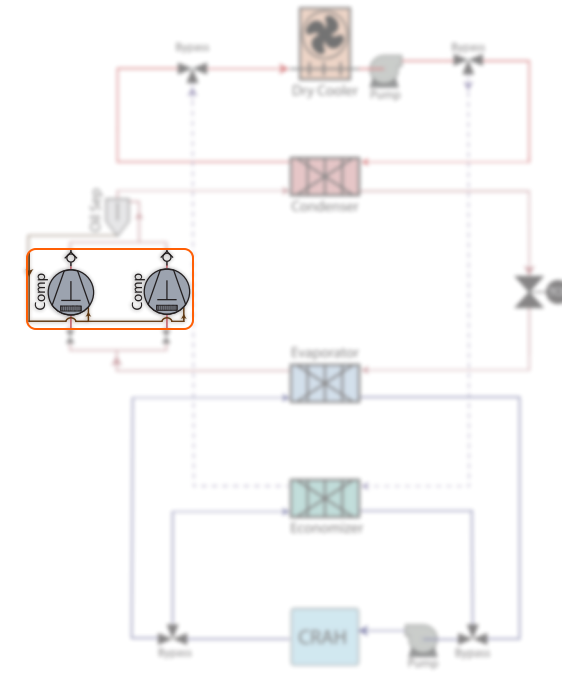
$$y = c_1 + c_2 t_o + c_3 t_c + c_4 t_o^2 + c_5 t_o t_c + c_6 t_c^2 + c_7 t_o^3 + c_8 t_c t_o^2 + c_9 t_o t_c^2 + c_{10} t_c^3$$

AHRI-540 / EN 12900 — separate coefficient sets for $\dot{Q}$, P and $\dot{m}$

$$\dot{Q}(N) = (N/N_0) \cdot \dot{Q}_0 \quad P(N) = (N/N_0) \cdot P_0$$

VFD model — capacity and power scale $\approx$ linearly with speed for a recip, so COP is speed-invariant

SELECTED MODEL → **2 × Bitzer 4FEP-35Z · semi-hermetic recip · VFD**



Compressors — location in the system



Bitzer ECOLINE semi-hermetic reciprocating, R-290-rated

Expansion Valve Selection and Modeling

- One EEV feeds the **full 150 kW** shared circuit, holding target evaporator superheat
- Sized by the incompressible-liquid orifice equation: **$A \approx 27.1 \text{ mm}^2 \rightarrow d \approx 5.9 \text{ mm}$** ($C_d = 0.65$)
- **One valve covers the full 5:1 turndown: 82.6% open at 150 kW $\rightarrow$ 24.3% at 30 kW** — always above the ~15% floor where EEV control degrades
- **Modeling: `calc_orifice()`** inverts the orifice equation at each load — required area over full area gives the opening fraction tracked across the turndown

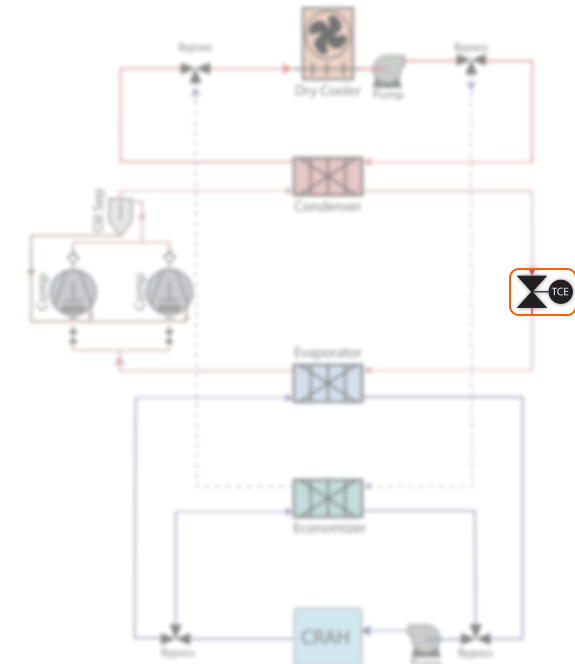
$$\dot{m}_r = C_d \cdot A_{\text{orifice}} \cdot \sqrt{2\rho_{\text{in}} \Delta P} \Rightarrow A_{\text{orifice}} = \frac{\dot{m}_r}{C_d \sqrt{2\rho_{\text{in}} \Delta P}}$$

Incompressible-liquid orifice equation, inverted for the required flow area

$$\text{opening} = A_{\text{req}}(\text{load}) / A_{\text{orifice}} \rightarrow 82.6\% \text{ at } 150 \text{ kW} \cdot 24.3\% \text{ at } 30 \text{ kW}$$

Valve-authority check across the 5:1 turndown (controls chapter)

SELECTED MODEL $\rightarrow$ **Emerson EX class · single valve**



Expansion valve — location in the system



Electronic expansion valve, stepper-driven

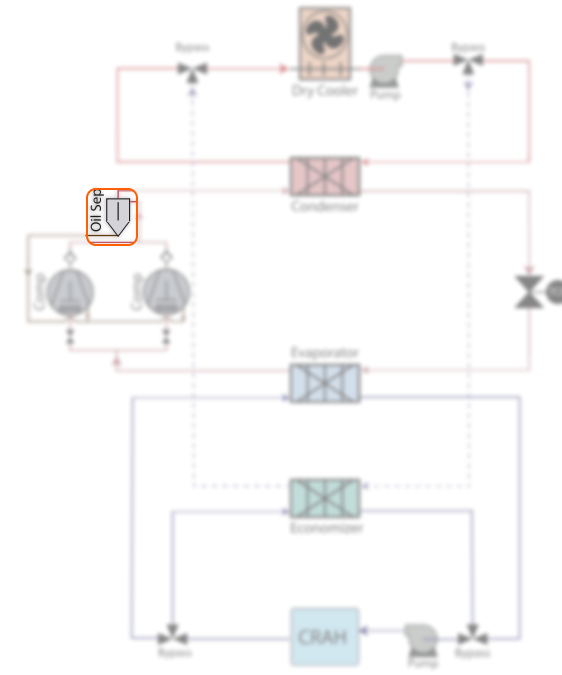
Oil Separator Selection and Modeling

- Returns oil from the common discharge to the crankcases — critical with **two parallel compressors** sharing one circuit
- Selected from Bitzer's chart by **maximum suction volume flow** — staying in the single-vendor Bitzer ecosystem
- 18 L oil charge; permitted for R-290 per Bitzer DB-300-9 / DP-500-2
- **Modeling:** the compressor map's mass flow and suction density give the worst-case volume flow — no thermal model needed, selection is a chart-band lookup

$$\dot{V}_{\text{suction}} = \dot{m}_r / \rho_{\text{suction}} \cdot 2 \text{ units} \rightarrow 203.6 \text{ m}^3/\text{h} \rightarrow \text{OA1954}$$

Bitzer quick-selection chart: sized on combined suction volume flow

SELECTED MODEL → **Bitzer OA1954 · primary · A3-rated · 28 bar**



Oil separator — location in the system



Bitzer OA-series primary oil separator

Dry Cooler Selection and Modeling

- **Duty:** 670 MBH at 42/38 °C glycol · 35 °C air — the **3 K approach** that anchors $t_c = 45$ °C and protects COP
- **Modeling:** cross-flow ϵ -NTU — sizing runs it **backward** (root-find NTU $\rightarrow$ UA = 73.7 kW/K), off-design runs it **forward** holding UA
- **No coil geometry from the vendor** $\rightarrow$ ΔP by **vendor-anchored scaling**: in the Blasius ratio, L, D and circuit count cancel — one rated point (55.2 kPa) scales to any condition (23.2 kPa at free cooling)
- **Variable-speed EC fans** (6 × 0.91 m, 60 dB(A)) — cube-law fan power in the annual model

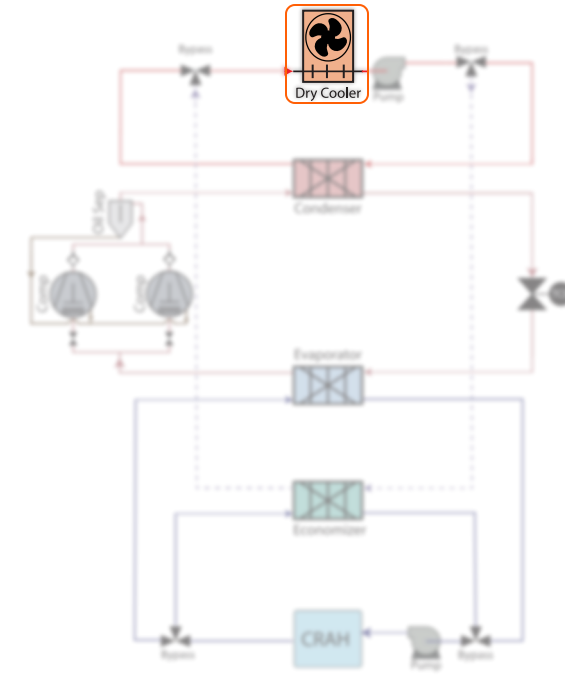
$$\epsilon = 1 - \exp \left\{ (1/C_r) NTU^{0.22} [\exp(-C_r NTU^{0.78}) - 1] \right\} \quad UA = NTU \cdot C_{\min}$$

Cross-flow ϵ -NTU (both streams unmixed) — NTU found by numerical root-finding

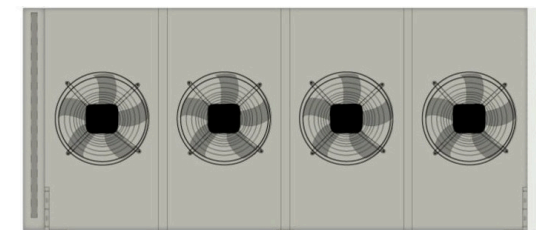
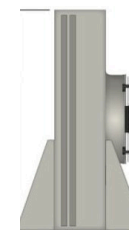
$$\Delta P_2 / \Delta P_1 = (\rho_2 / \rho_1)^{0.8} (\dot{m}_2 \rho_1 / \dot{m}_1 \rho_2)^{1.8} (\mu_2 / \mu_1)^{0.2}$$

Vendor-anchored ΔP scaling — geometry cancels; needs one rated point + CoolProp only

SELECTED MODEL $\rightarrow$ **Kelvion ULF-PA106K4V-091F095 · 6 EC fans**



Dry cooler — location in the system



Kelvion flat dry cooler, variable-speed EC fans

Pump Selection and Modeling

- **Glycol loop — 2.5AC:** 41.4 m³/h @ 14.2 m (11.64 kg/s of 30% PG), ≈2.57 kW shaft
- **Chilled-water loop — 1.5AD:** 7.0 in trim, 1.5 hp, serving the CRAH circuit
- **Digitized for modeling:** vendor curve sheets traced per impeller trim (5.0–7.0 in) and least-squares fit — the model runs on the fitted curves, not catalog points
- Both on **VFDs** — pump parasitics track flow in the annual PUE model instead of running flat-out
- **Modeling:** fitted $H(Q)$ intersected with the loop resistance curve; the digitized efficiency map turns duty into shaft power for the 8,760-h PUE model

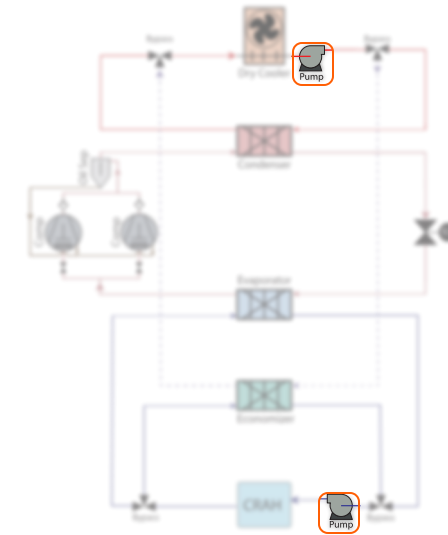
$$H(Q) = c_2 Q^2 + c_1 Q + c_0 \quad P_{\text{shaft}} = \rho g Q H / \eta_{\text{pump}}$$

Quadratic fit to each digitized trim curve · trim chosen where $H(Q)$ clears the margined duty

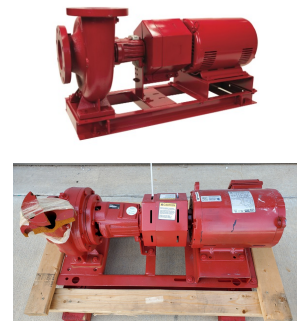
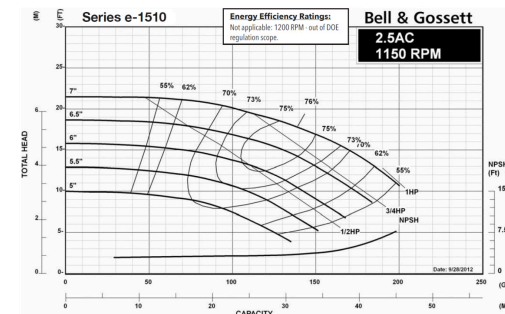
$$Q \propto N \quad H \propto N^2 \quad P \propto N^3$$

Affinity laws — VFD part-load power in the annual model

SELECTED MODEL → Bell & Gossett e-1510 · both VFD · 1750 RPM

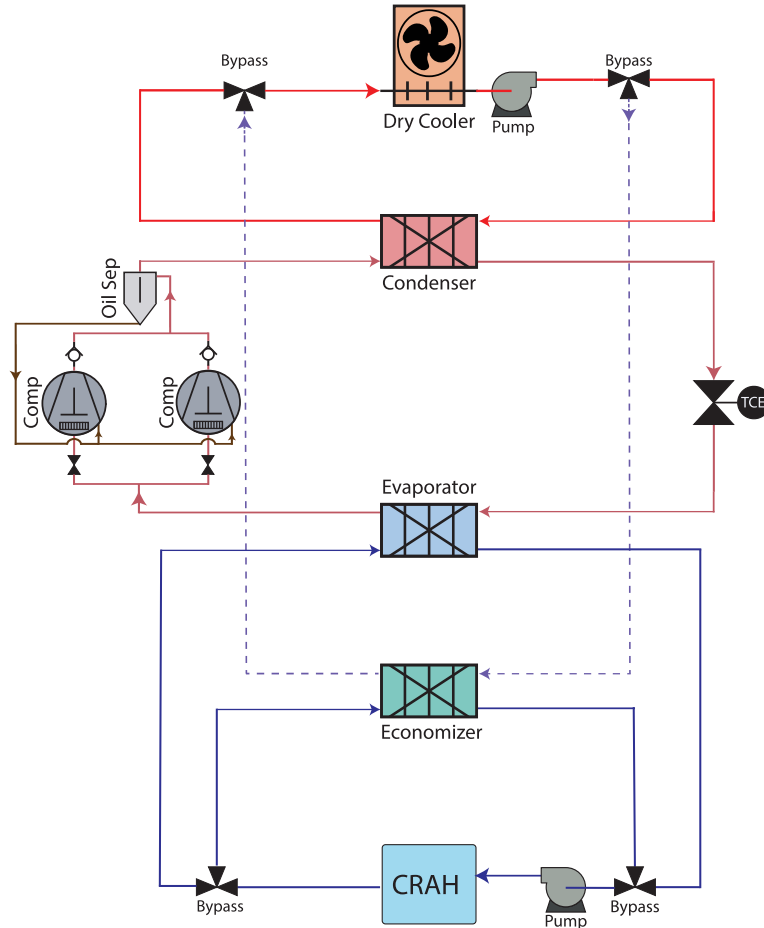


Pumps — glycol (top) and chilled-water (bottom) locations



B&G e-1510 pumps · 2.5AC curve sheet digitized for $H(Q)$ & efficiency → shaft power

Full-Load COP — Whole Plant, Every Component Modeled



System P&ID — every block is in the performance model

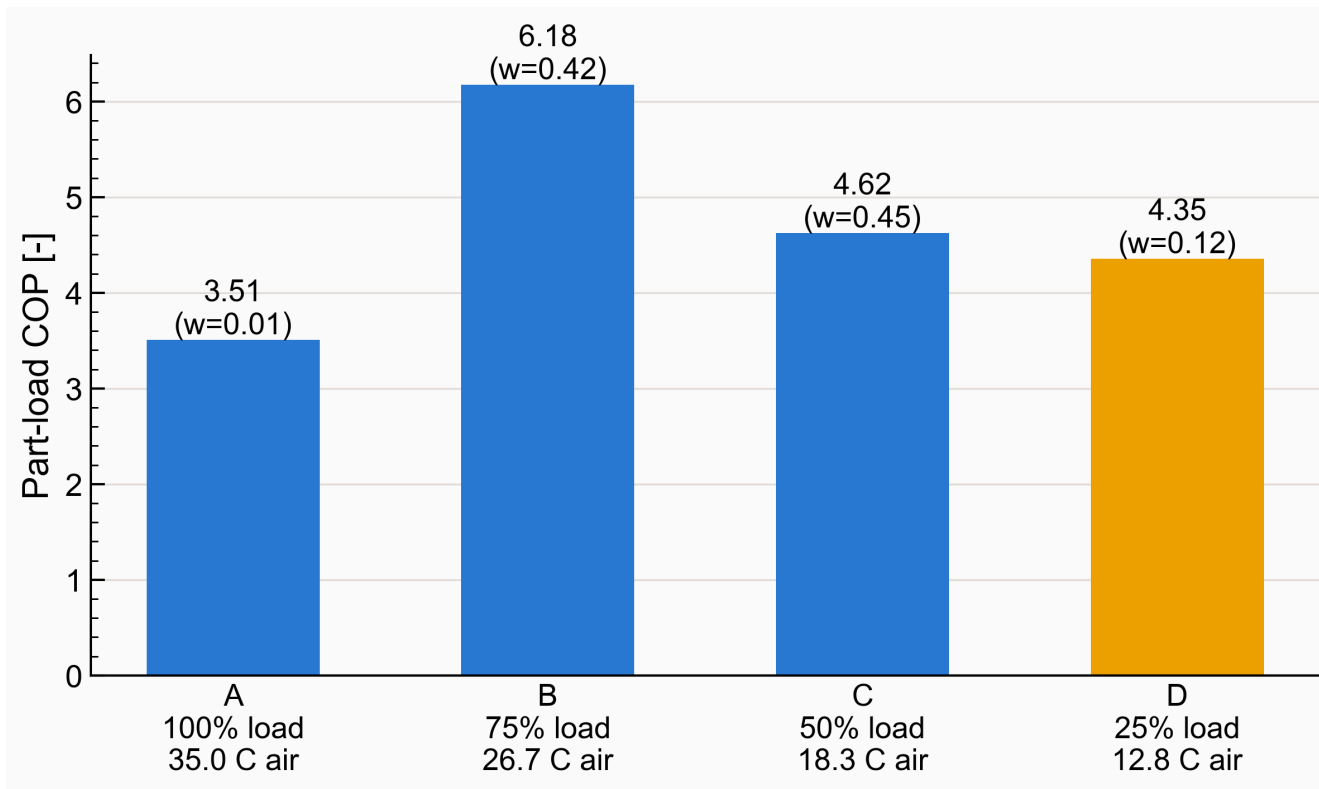
FULL-LOAD COP · WHOLE PLANT · DESIGN POINT

3.59

Every component is modeled, not lumped — compressor (Bitzer AHRI-540 map), dry-cooler EC fans, glycol & chilled-water pumps, economizer and dry cooler — with **all parasitics included** at the design point.

IPLV & the 5:1 Turndown

The VFDs modulate capacity down to the 25% rating point — no hot-gas bypass — so part-load COP climbs as the condenser gets colder.



IPLV.IP part-load points A–D per AHRI 550/590-2023

INTEGRATED PART LOAD VALUE

5.23

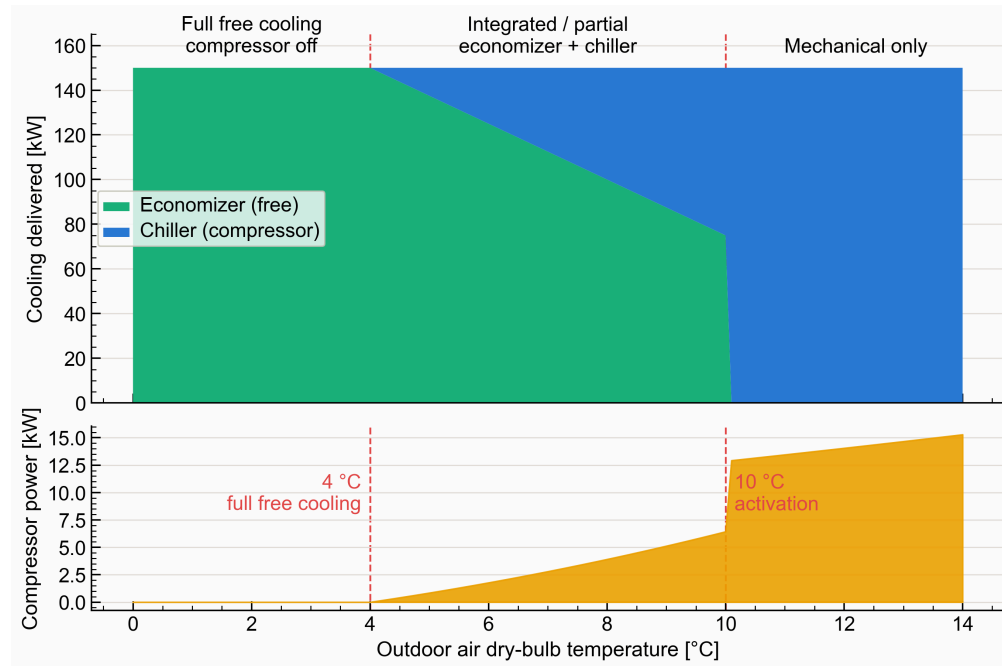
≥ 5.0 target met with a 4.6% margin. Speed modulation is also what makes the 30 kW minimum load affordable.

Results — Free Cooling

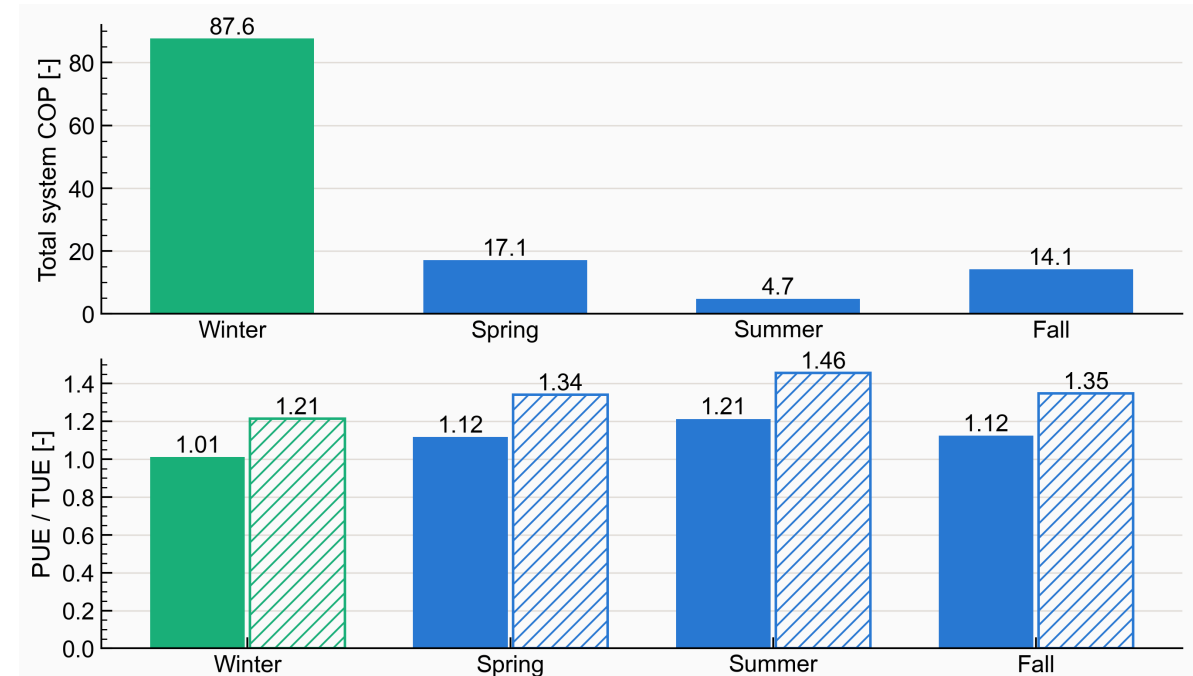


The Economizer Pays Back Every Winter Hour

Across the economizer band the compressor's share of the load collapses; over the Champaign TMY3 year, **~40% of all cooling** is delivered with almost no compressor work — winter PUE approaches **1.01**.



Load split & compressor power across the ASHRAE 90.1 economizer envelope



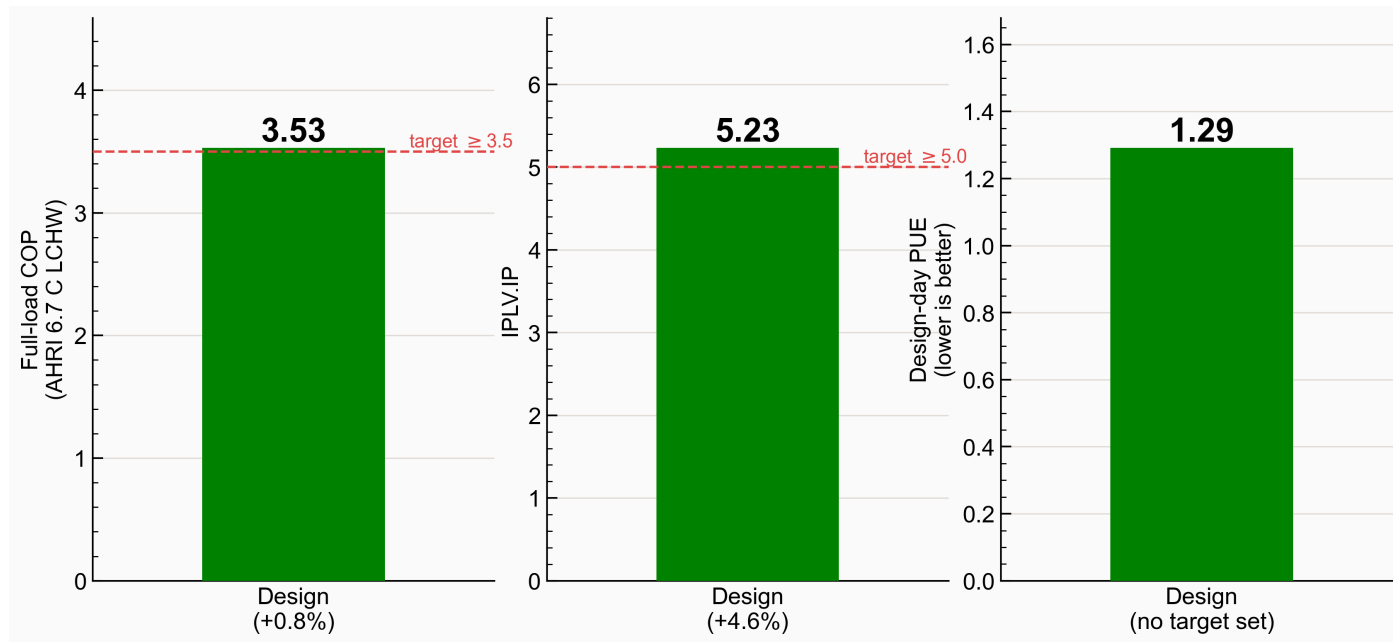
Seasonal-average total-system COP, PUE and TUE

Results — PUE & Compliance



Both Mandatory Targets Cleared

The compressor bank dominates design-day facility power; fans and pumps stay small. The design-day PUE is the worst case — the annual average is far lower.



The design against the two mandatory efficiency targets

DESIGN-DAY FACILITY PUE

1.29

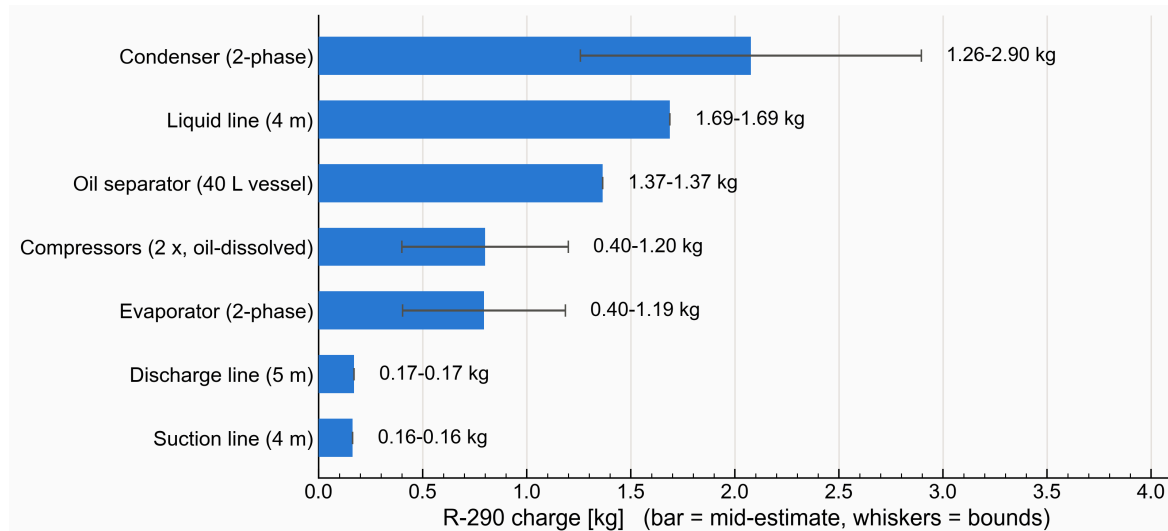
COP **3.53** ✓ · IPLV **5.23** ✓ — the COP margin is thin, the IPLV margin comfortable. Both computed from the same AHRI-540 map and fan/pump models.

Safety — A3 Charge Analysis

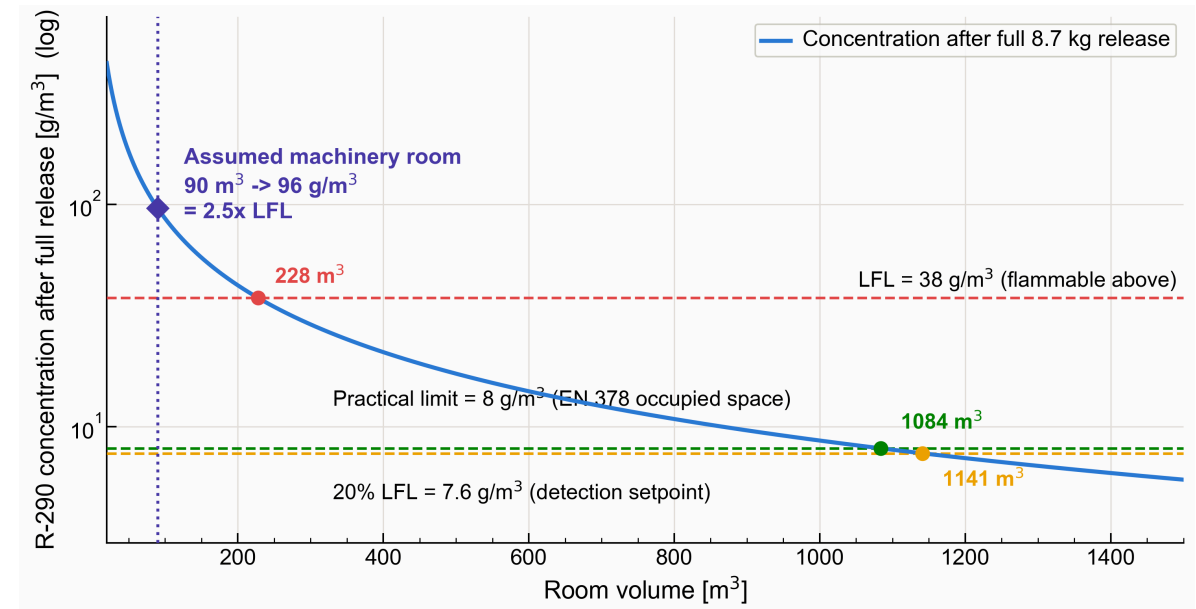


Small Charge, Kept Out of the Hall

- Total R-290 charge is **5.5–8.7 kg** — minimized (no receiver, 4 m liquid line) and kept out of the hall by the glycol loop.
- Dilution alone **won't keep the plant room below LFL** — a full release reaches $\sim 2.5\times$ LFL.
- Safety rests on an **EN 378 machinery room**: 20% LFL gas detection + emergency ventilation.



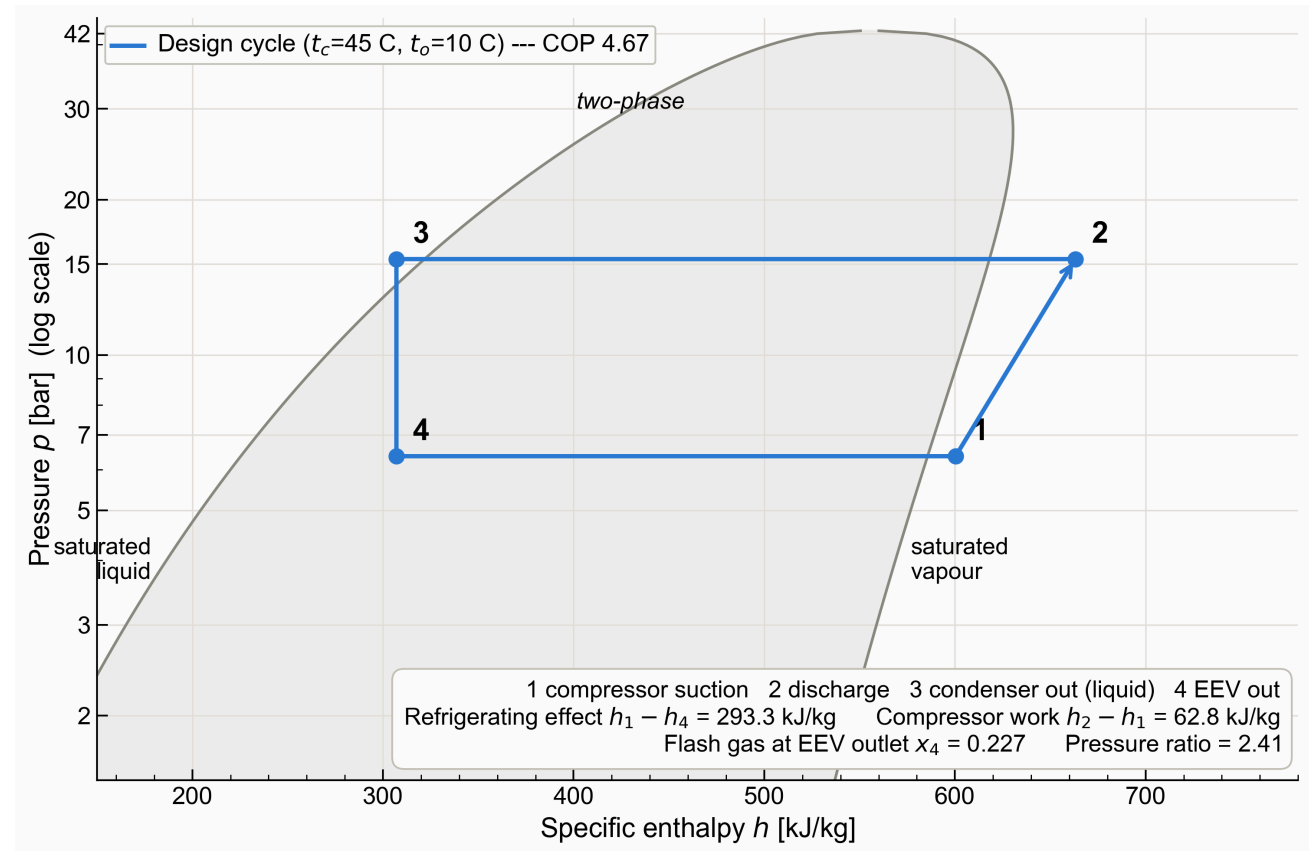
Charge inventory at the design point — condenser holdup dominates



Room concentration after a full release vs. the safety limit

Both Targets Met with an A3-Safe Architecture

- **R-290 three-loop chiller** — propane never enters the data hall
- Every major component **vendor-selected** on real data (Bitzer, Kelvion, B&G)
- Both mandates met: **COP 3.53 · IPLV 5.23 ·** design-day PUE 1.29
- Champaign winters do **≈40% of the cooling for free** via the economizer



The designed cycle — R-290 p-h chart at 150 kW / 35 °C



Air Conditioning and Refrigeration Center

Thank you!

Questions & discussion welcome

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IRCC 2026 · Project #3 — 150 kW R-290 chiller · COP 3.53 · IPLV 5.23 · PUE 1.29